Conformal Multi-Agent Topologies: Anytime-Valid Communication Design with Distribution-Free Answer Guarantees

Research Proposal — Technical Report

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Abstract

Recent LLM-based multi-agent systems (MAS) rely on *communication topologies* — who talks to whom, how often, and about what — to unlock collective intelligence. State-of-the-art topology designers (G-Designer, AgentPrune, PEAR, DySCo, AMAS) are either *one-shot* (they commit a topology before any LLM response is observed) or use uncertainty only as an *ad-hoc weight* on messages (DebUnc). None provides a distribution-free finite-sample guarantee on the correctness of the final answer, and none couples topology design with a principled stopping rule — a gap explicitly flagged in the May 2026 survey on RL for MAS. We propose **ConfTopo**, a framework that (i) treats each inter-agent message as a *conformal prediction set* over verifiable claims, (ii) grows the communication topology round-by-round using an *e-process* over agent agreement, and (iii) returns an answer with provable end-to-end miscoverage $\leq \alpha$ under a calibration-exchangeability assumption on tasks. The core technical contribution is a *compositional coverage theorem* that lifts per-message CP guarantees to a graph-level guarantee on the answer at the sink node, together with an anytime-valid topology-growth policy that provably preserves this guarantee at any stopping time. We detail the algorithm, situate it precisely against 2024–2026 work on MAS topology design and conformal LLMs, and lay out a NeurIPS/ICML-scale experimental plan on MMLU, HumanEval, MATH, MedQA, and MuSR, with baselines including G-Designer, AgentPrune, PEAR, DySCo, DebUnc, and AgentDropout.

1 Introduction

LLM-based multi-agent systems now routinely outperform single-agent baselines on reasoning, coding and QA benchmarks by orchestrating multiple LLM instances over a directed communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The design of $\mathcal{E}$ — the *topology* — is a first-order determinant of both accuracy and token cost. G-Designer (ICML 2025) established the current paradigm: a variational graph auto-encoder consumes a task embedding plus per-agent role features and *decodes* a task-adaptive topology in a single forward pass. Follow-ups add sparsity priors (AgentPrune / Cut-the-Crap), one-shot pruning masks, or per-round adaptive routing (PEAR, DySCo, AMAS).

Two critical properties are still *missing* from the entire literature:

1. **Response-conditioned topology.** G-Designer designs $\mathcal{G}$ *before* any agent speaks; it never conditions on what agents actually say or on *how confident* they are. DebUnc (EMNLP 2025) injects uncertainty metrics but only as attention weights on tokens — the topology itself is fixed. Even the most recent dynamic methods (PEAR, DySCo) use uncertainty *heuristically* for edge scoring, with no formal guarantee.
2. **Statistical guarantees on the answer.** No MAS topology paper can state a claim of the form "for any task from a fixed distribution, the returned answer is correct with probability at least $1-\alpha$." Cost–accuracy Pareto curves are reported empirically but never certified.

Conformal prediction (CP) is the standard machinery for distribution-free finite-sample guarantees; it has been ported to LLMs (Mohri & Hashimoto's conformal factuality; Cherian, Gibbs, Candès on enhanced CP; Quach et al. on conformal language modeling), to GNNs (CF-GNN, GraphLCP), and to federated multi-agent settings (Personalized Federated Weighted CP, 2606.00717). Yet no existing work uses CP to *design a communication topology.* The naive combination — "run G-Designer, then wrap the output with a conformal predictor" — does not work, because the object we care about (the final answer at the sink node) is produced by a *stateful, sequential, graph-structured process* whose messages are not exchangeable across rounds.

Contribution. We propose **ConfTopo**, the first framework that jointly (a) selects which edges fire, (b) filters what each edge transmits, and (c) decides when to stop, under a single end-to-end coverage guarantee. Concretely:

1. *Conformal message gating.* Each agent's outgoing message is a conformal prediction set over verifiable sub-claims; edges that would forward oversized sets are muted, so agents *learn to abstain* without any fine-tuning.
2. *E-process topology growth.* Between rounds, a topology-growth policy adds edges/agents only when a nonnegative-martingale test statistic fails to certify convergence, giving *anytime-valid* error control.
3. *Compositional coverage theorem.* We prove that if each message satisfies a per-edge $(1-\alpha_e)$-coverage and the sink-node aggregator is monotone, then the final answer satisfies end-to-end miscoverage $\leq \alpha$, where α is an explicit function of the α_e's and the graph structure.

To our knowledge this is the first MAS framework whose central claim is a *statistical* statement about the answer, not just an empirical cost–accuracy curve.

2 Related Work and Positioning

2.1 Topology design for LLM multi-agent systems

Static / one-shot designers. G-Designer [1] uses a VGAE that consumes a task-specific virtual node and agent role embeddings to produce a topology; the design is committed before any agent responds. AgentPrune / Cut-the-Crap [2] learns a low-rank spatio-temporal mask that one-shot-prunes redundant edges. AgentDropout [3] eliminates entire agents. Assemble-Your-Crew [4] autoregressively generates topologies. All of these fix $\mathcal{E}$ before observing responses.

Adaptive routers. DySCo (2606.01828) and PEAR (2606.20621) select sparse topologies at each debate round from agent states, using trust or divergence heuristics. AMAS (EMNLP 2025 Industry Track) adapts the topology per task. These methods react to agent output but supply no coverage guarantee; their claims are cost/accuracy Pareto curves only.

Uncertainty-aware communication. DebUnc [5] conveys per-agent uncertainty (mean token entropy, TokenSAR) via a modified attention mechanism or textual prompt, biasing peers toward more confident agents. DWC-MAD (KSEM 2026) adds a longitudinal accuracy weighting. Uncertainty here is a *signal for weighting*, never for topology or stopping, and never with a distribution-free guarantee.

2.2 Conformal prediction for LLMs and graphs

Recent CP-for-LLM work provides scalar confidence on generations (SConU at ACL 2025; TECP in Mathematics 2025), prediction sets for MCQ [10], factuality guarantees [7], black-box conformal LMs [9], and enhanced CP for open-ended generation [8]. CF-GNN [6] extends split CP to node-classification GNNs with a topology-aware correction; GraphLCP (2605.08074) adds structure-aware localized weights; non-exchangeable temporal CP on graphs appears in 2507.02151. Multi-agent CP work targets *federated calibration* under distribution shift (2606.00717), not communication design. "From Debate to Decision: Conformal Social Choice" (2604.07667) applies CP to *voting* over agents' outputs but treats the topology as given.

2.3 The precise gap we close

Existing work either designs topologies *without* uncertainty and without guarantees (G-Designer family), uses uncertainty as an *unguaranteed* heuristic (DebUnc, PEAR, DySCo), or applies CP to *fixed* MAS pipelines (conformal social choice, federated CP). **No prior work uses CP to *design* a topology, and none provides end-to-end coverage on the final MAS answer.** ConfTopo occupies exactly this cell of the matrix and is not reducible to any composition of the above because (i) the exchangeability assumption breaks across debate rounds and (ii) the object being covered is a graph-aggregated, potentially free-form answer.

3 Problem Formulation

Setup. A task $T \sim P_{\mathcal{T}}$ is drawn from a distribution over tasks (e.g. MMLU-style questions). We have a pool of N LLM agents $\mathcal{V} = \{v_1, \dots, v_N\}$ with fixed roles/prompts. A *topology* at round t is a directed graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ over some subset of active nodes. Communication proceeds in rounds $t = 1, \dots, R$; at each round each active agent v_i receives messages from its in-neighbors, generates a message m_i^t, and forwards it to its out-neighbors. After the final round, an aggregator Φ at a sink node produces an answer $\hat{y} = \Phi(\{m_i^t\}_{i,t})$.

Miscoverage. Given ground-truth answer $y^\star(T)$ and a task-appropriate correctness predicate $\mathbb{I}\{\hat{y} \equiv y^\star\}$, the *miscoverage rate* is

$$\mathrm{MisCov}(\pi) \;=\; \mathbb{P}_{T \sim P_{\mathcal{T}}}[\hat{y}(\pi, T) \neq y^\star(T)]\,, \tag{1}$$

where π is the full pipeline (topology-design policy + agents + aggregator). Our goal is a pipeline π satisfying $\mathrm{MisCov}(\pi) \leq \alpha$ for user-specified $\alpha \in (0, 1)$, while minimizing expected token cost $\mathbb{E}[C(\pi, T)]$.

Calibration exchangeability. We assume access to a calibration set $\{(T_i, y_i^\star)\}_{i=1}^n$ i.i.d. from $P_{\mathcal{T}}$. Following the standard CP recipe, we require calibration and test tasks to be exchangeable *across tasks*, but explicitly *not* across rounds within a task — messages within one task are highly dependent and this is where the methodological difficulty lies.

4 Method: ConfTopo

ConfTopo has three coupled components (Figure 1): (i) a per-message conformal filter, (ii) an anytime-valid topology-growth policy, and (iii) a monotone aggregator with a compositional coverage guarantee.

[Overview diagram, to be produced for final paper]

Left: seed topology $\mathcal{G}_0$. *Middle*: each edge carries a conformal prediction set $\mathcal{C}_{ij}^t$; edges whose set exceeds a size threshold are muted. *Right*: an e-process $M_t = \prod_{s \le t} e_s$ is updated from per-round agreement; when $M_t \ge 1/\alpha$ the sink returns $\hat{y}$, otherwise the growth policy adds edges/agents to $\mathcal{G}_{t+1}$ using a budgeted rule.

Figure 1: ConfTopo end-to-end. Conformal filters gate messages; e-process controls stopping; growth policy expands the topology.

4.1 Per-message conformal filtering

Let m_i^t denote the raw response of agent v_i at round t. We decompose m_i^t into a set of atomic *sub-claims* $\{c_{i,k}^t\}_k$, e.g. via an LLM-based claim extractor (as in conformal factuality). To each sub-claim we assign a *nonconformity score* $s(c_{i,k}^t \mid \text{context}) \in \mathbb{R}$; we instantiate s as one of:

- negative log-probability of the claim under agent v_i's LLM;
- semantic-entropy over paraphrases (Kuhn et al., 2023);
- cross-agent disagreement between v_i's claim and a probe agent.

Given a per-edge miscoverage budget α_e, split-CP on the calibration set produces a threshold $\hat{q}_e(\alpha_e)$, and the *message prediction set* that v_i forwards along edge (i, j) is

$$\mathcal{C}_{ij}^t \;=\; \left\{\, c_{i,k}^t \;:\; s(c_{i,k}^t) \le \hat{q}_e(\alpha_e) \right\}. \tag{2}$$

Gating. Given a size budget $B \in \mathbb{N}$, edge (i, j) *fires* iff $|\mathcal{C}_{ij}^t| \le B$; otherwise agent v_i is required to either (a) abstain from that edge, or (b) transmit only the sub-claim with the lowest nonconformity score plus an explicit "low-confidence" flag. This gives a learned abstention mechanism with no LLM fine-tuning, sitting entirely at the message layer.

Proposition 1 (Per-edge validity)**.** *Under exchangeability of calibration and test claims for edge (i, j), the prediction set $\mathcal{C}_{ij}^t$ satisfies $\mathbb{P}\left[c_{ij}^\star \in \mathcal{C}_{ij}^t\right] \ge 1 - \alpha_e$, where $c_{ij}^\star$ is the ground-truth relevant claim for the downstream computation at v_j.*

Proposition 1 is a direct application of split CP; the methodological content lies in composing many such guarantees.

4.2 Compositional coverage on the graph

Fix a DAG $\mathcal{G}_t$. Let ϕ_j be the aggregator at node v_j, which consumes its incoming message-sets and produces its own belief. Say ϕ_j is *monotone* if adding a covered claim to any input set can only weakly improve the correctness of ϕ_j's output. This holds, for example, when ϕ_j is an LLM prompted to answer given a union of claims (more true claims cannot hurt in expectation), and when hallucination is bounded by a separate factuality-CP layer.

Theorem 2 (Compositional coverage; sketch)**.** *Let $\mathcal{G}_t$ be a DAG with sink $v^\star$ and let paths$(v^\star)$ denote the set of source-to-sink paths. Assume Proposition 1 holds edge-wise with budgets $\{\alpha_e\}_{e \in \mathcal{E}_t}$ and the sink aggregator $\phi_{v^\star}$ is monotone. Then the sink's answer is $y^\star$-covered with probability at least*

$$1 - \sum_{p \in paths(v^\star)} \prod_{e \in p} \alpha_e \;\ge\; 1 - |\mathcal{E}_t| \cdot \max_e \alpha_e. \tag{3}$$

Proof sketch. By the monotone-aggregator assumption, a path-covered claim suffices to make the sink correct. The probability that *every* path fails is bounded by the product of edge miscoverages along that path (by independence of the CP calibration draws for each edge — an assumption we make precise and defend in §4.5). A union bound over paths yields the stated rate. A tighter form uses Fréchet–Hoeffding inequalities when edge-CPs share calibration data, at the cost of a factor depending on the graph diameter. □

Remark 3. Theorem 2 inverts the topology-design objective. Given a *target* miscoverage α and a graph $\mathcal{G}_t$, we can *solve* for the largest edge budgets α_e compatible with α; larger α_e means smaller conformal sets and more edge gating, hence lower token cost. This makes ConfTopo a *coverage-constrained sparsifier*.

4.3 Anytime-valid topology growth

Rounds are not exchangeable, so we cannot apply split CP across rounds directly. Instead we build a nonnegative supermartingale (e-process) $\{M_t\}_{t\geq 0}$ under the null hypothesis H_0: "the current answer $\hat{y}_t$ is incorrect."

Constructing the e-process. At round t let $a_i^t \in [0,1]$ be a bounded, mean-known-under-H_0 statistic derived from agent agreement with the leading answer (e.g. fraction of active agents whose top-1 answer equals $\hat{y}_t$, calibrated so $\mathbb{E}_{H_0}[a_i^t] = \mu_0$). Define round-level e-values

$$e_t \;=\; \exp(\lambda_t(a_t - \mu_0) - \psi(\lambda_t))\,, \qquad M_t \;=\; \prod_{s\leq t} e_s, \tag{4}$$

where a_t is the round-t agreement, $\lambda_t \in [0, 1/(1-\mu_0)]$ is a betting parameter chosen predictably from history, and ψ is the cumulant generating function on $[0,1]$. By construction $\{M_t\}$ is a nonnegative supermartingale under H_0, so by Ville's inequality, $\mathbb{P}_{H_0}[\exists t : M_t \geq 1/\alpha] \leq \alpha$ for any stopping rule.

Growth policy. If $M_t < 1/\alpha$ we grow $\mathcal{G}_{t+1}$: pick the *single* edge (or fresh agent) whose expected contribution to M_{t+1} is largest under a cheap surrogate model, subject to a token budget. Because $\{M_t\}$ is anytime-valid, the growth policy may depend adaptively on everything observed so far — a property that split CP lacks and that is essential for MAS.

Proposition 4 (Anytime coverage under growth)**.** *Let $\tau = \inf\{t : M_t \geq 1/\alpha\}$ (with $\tau = \infty$ if never). Then $\mathbb{P}[\hat{y}_\tau \neq y^\star \wedge \tau < \infty] \leq \alpha$.*

Combined with Theorem 2 applied at the stopping time, this yields end-to-end miscoverage $\leq \alpha$ regardless of the topology's final size or the number of rounds — the property currently missing from the entire MAS literature.

4.4 Full algorithm

Algorithm 1 ConfTopo: Conformal Multi-Agent Topology

Require: task T, agent pool $\mathcal{V}$, seed graph $\mathcal{G}_0$, coverage α, edge budget α_e, size bound B, token budget $C_{\max}$

Fit per-edge CP thresholds $\hat{q}_e(\alpha_e)$ on calibration data
$t \leftarrow 0$, $M_0 \leftarrow 1$
while $M_t < 1/\alpha$ **and** tokens-used $< C_{\max}$ **do**
  $t \leftarrow t + 1$
  for each active edge $(i, j) \in \mathcal{E}_t$ **do**
    v_i generates response m_i^t; extract sub-claims $\{c_{i,k}^t\}$
    Compute nonconformity scores $s(c_{i,k}^t)$
    Form $\mathcal{C}_{ij}^t = \{c_{i,k}^t : s(\cdot) \leq \hat{q}_e(\alpha_e)\}$
    if $|\mathcal{C}_{ij}^t| \leq B$ **then** forward $\mathcal{C}_{ij}^t$ to v_j
    else abstain or forward best-claim + flag
    end if
  end for
  Aggregate at sink: $\hat{y}_t \leftarrow \phi_{v^\star}$(received sets)
  Compute round agreement a_t; update λ_t predictably
  $M_t \leftarrow M_{t-1} \cdot \exp(\lambda_t(a_t - \mu_0) - \psi(\lambda_t))$
  if $M_t < 1/\alpha$ **then**
    $\mathcal{G}_{t+1} \leftarrow \mathcal{G}_t \cup \{$edge maximizing surrogate $\Delta M\}$
  end if
end while
return $\hat{y}_t$

4.5 Assumptions and where they might break

The two load-bearing assumptions are (A1) calibration-exchangeability of per-edge claims and (A2) monotone aggregation at the sink. (A1) is standard for split CP but requires per-edge calibration data — we discuss weight-sharing across edges below. (A2) can fail if an aggregator LLM is *misled* by additional true claims (e.g. Simpson's-paradox-style prompts). We address (A2) empirically via a factuality-CP sanity layer on the aggregator and theoretically via a bounded-loss weakening (Corollary 5).

Corollary 5 (Non-monotone aggregator)**.** *If the sink aggregator has a bounded degradation rate* $|\phi(S \cup \{c\}) - \phi(S)| \leq \eta$ *per added claim, then miscoverage is bounded by* $\alpha + \eta \cdot \mathbb{E}[|\textit{aggregated set}|]$.

5 Experimental Plan

We describe an ICML/NeurIPS-scale experimental program. We separate *primary* experiments (needed for the main result) from *ablations* and *analyses*.

5.1 Benchmarks

To ensure comparability with G-Designer, AgentPrune, PEAR, DySCo we cover the standard suite plus two harder settings:

- **MMLU** (multi-choice, 57 subject subsets — allows per-subject group-conditional analysis).
- **HumanEval** and **MBPP** (code, pass@1).
- **MATH** (open-ended reasoning, exact-match).
- **MedQA / MedMCQA** (high-stakes; coverage guarantees matter most here).
- **MuSR** (multi-step soft reasoning; stress-tests aggregator monotonicity).

5.2 Models

Three model families to test generality: GPT-4-class (proprietary), Llama-3.1-70B-Instruct (open weights), and Qwen-2.5-72B. Each agent in $\mathcal{V}$ instantiated identically per condition to isolate topology effects.

5.3 Baselines

- **Single agent** (CoT and CoT-SC).
- **Fixed topologies**: chain, star, complete, GPTSwarm.
- **Learned static**: G-Designer, AgentPrune, AgentDropout.
- **Dynamic**: PEAR, DySCo, AMAS.
- **Uncertainty-aware**: DebUnc (both attention and prompt variants).

For fairness, all baselines are given the same *token budget* $C_{\max}$ and the same agent pool.

5.4 Metrics

- **Empirical miscoverage** vs. target α (for ConfTopo, this is the guarantee; for baselines, it is descriptive).
- **Accuracy** at matched cost.
- **Token cost** at matched accuracy (Pareto curves).
- **Set size** $|\mathcal{C}_{ij}^t|$ and abstention rate.
- **Stopping-time distribution** $\mathbb{P}[\tau = t]$ for ConfTopo.
- **Robustness** to adversarial agents (following G-Designer's protocol, one adversarial agent argues an incorrect answer).

5.5 Main claims to test

H1. ConfTopo achieves empirical miscoverage $\leq \alpha$ across all benchmarks and $\alpha \in \{0.05, 0.1, 0.2\}$; baselines do not.

H2. At matched miscoverage, ConfTopo dominates the cost–accuracy Pareto frontier of all baselines by $\geq 2\times$ tokens on average.

H3. Under adversarial attack (1 adversarial agent), ConfTopo's accuracy drop is $< 1\%$ (G-Designer reports 0.3%; PEAR reports larger drops).

H4. The stopping time τ correlates with instance-level difficulty (higher for MATH-Hard, lower for MMLU-High-School), giving *adaptive compute* for free.

5.6 Ablations

- Remove conformal filtering (keep e-process only): tests whether per-edge sets matter or e-process alone suffices.
- Remove e-process (fixed number of rounds): tests whether anytime growth matters vs. conservative fixed-R.
- Nonconformity choice: negative log-prob vs. semantic entropy vs. cross-agent disagreement.
- Split-CP vs. weighted CP under task-family shift.
- Sink aggregator: majority vote vs. LLM aggregator vs. conformal majority (conformal social choice, 2604.07667).

5.7 Analyses

- *Coverage-per-group*: does ConfTopo maintain $\leq \alpha$ miscoverage *per MMLU subject*? If not, add a multi-valid layer.
- *Calibration size sensitivity*: how many calibration tasks does ConfTopo need to hit target coverage within $\pm 1\%$?
- *Failure modes*: cases where Theorem 2 is tight (many parallel paths saturating the union bound) vs. loose.
- *Compute-adaptive scaling law*: plot $\mathbb{E}[\tau]$ vs. task difficulty (measured by single-agent CoT accuracy) and fit a scaling relation, contrasted with fixed-R methods.

6 Positioning: what makes this A*-plausible

An ICML/NeurIPS Oral or Spotlight in this area typically requires (i) a new formulation that reframes an existing problem, (ii) a non-trivial theorem, and (iii) empirical dominance at matched cost. We believe ConfTopo scores on all three:

- **Reformulation.** MAS topology design is recast as a *coverage-constrained sparsification* problem, with cost as the objective and miscoverage as the constraint. This inverts the G-Designer objective (design for accuracy, hope cost is low) and gives a principled axis (coverage) that current methods cannot even report.
- **Theory.** The compositional coverage theorem and anytime-valid growth guarantee are new to MAS. Neither CF-GNN (node-level CP) nor conformal social choice (voting-level CP) nor federated CP (across-agent calibration) provides an edge-level $\rightarrow$ answer-level bridge for sequential, non-exchangeable multi-round MAS.
- **Empirical bite.** Because the coverage guarantee holds for *any* stopping time, ConfTopo can spend far fewer tokens on easy tasks and more on hard ones without sacrificing correctness — exactly the property MAS practitioners want but currently obtain only via task-specific heuristics.

The most likely reviewer objection is that split-CP's exchangeability across tasks is fragile under domain shift. We plan to address this directly with a weighted-CP variant (following Tibshirani et al. 2019 and its 2024–2025 LLM adaptations) and by reporting per-family coverage. A secondary concern is calibration cost; we quantify this in the calibration-sensitivity ablation.

7 Risks, Limitations, and Broader Impact

Risks. (1) The monotone-aggregator assumption may fail on strongly leading-question sinks — Corollary 5 bounds the degradation but does not remove it. (2) Nonconformity scores from log-probabilities may be poorly calibrated for RLHF'd models; semantic entropy is more expensive. (3) Multi-round e-processes can be conservative if λ_t is chosen poorly; predictable betting strategies (ONS-betting) mitigate this.

Limitations. We provide *marginal* coverage; per-group coverage requires the multi-valid extension (§4 ablations). Adversarial robustness is empirical, not certified — a certified extension is a natural follow-up.

Broader impact. Providing distribution-free correctness guarantees on MAS answers is directly relevant to high-stakes deployments (medical, legal, safety-critical code). We caution that guarantees are conditional on the calibration exchangeability assumption; deployed systems must monitor for drift.

8 Conclusion

We proposed ConfTopo, a framework that designs LLM-MAS communication topologies under a distribution-free finite-sample guarantee on the answer, by combining per-message conformal filtering with anytime-valid topology growth via e-processes. The framework resolves the two central gaps in the current MAS-topology literature: it conditions on LLM responses (unlike G-Designer) and it provides formal coverage (unlike DebUnc, PEAR, DySCo). We have specified the algorithm, sketched the two key theorems, and laid out a NeurIPS/ICML-scale experimental plan with baselines drawn from the 2024–2026 state of the art.

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